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Instructions

- Show all steps for full credit. You may type your responses or submit clear handwritten work.
- Submit a single PDF file.
- Total points: 20

Problem 1: Prediction Using Coefficients (3 pts)

Given the following multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

with estimated coefficients: $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = 3$, and $\hat{\beta}_2 = -1.5$.

Predict the value of Y when $X_1 = 4$ and $X_2 = 2$.

$Y = 2 + 3X_1 - 1.5X_2 + \epsilon$

$Y = 2 + 3(4) - 1.5(2) + \epsilon$

$\boxed{Y = 11 + \epsilon \text{ or } Y = 11}$

Problem 2: Adjusted R^2 Calculation (4 pts)

A regression model is fitted on a dataset with $n = 100$ observations and $p = 3$ predictors. The coefficient of determination is $R^2 = 0.8$.

Compute the Adjusted R^2 and show each step in your calculation.

$$\text{Adjusted } R^2 = 1 - \frac{(1-R^2)(n-1)}{n-p-1}$$

$$\text{Adj } R^2 = 1 - \frac{(1-.8)(99)}{100-3-1} = 1 - \left(\frac{19.8}{96}\right) = .79375$$

$\boxed{\text{Adj } R^2 = .79375}$

Problem 3: Significance of Predictors (5 pts)

A regression model includes predictors X_1, X_2, and X_3 with the following coefficient estimates and standard errors:

No sample size given, so I'm going off of the notes. If I did have sample size, I would use z if $n > 30$, and t if $n \leq 30$.

$$\hat{\beta}_1 = 5, \quad SE(\hat{\beta}_1) = 1$$
$$\hat{\beta}_2 = -3, \quad SE(\hat{\beta}_2) = 0.5$$
$$\hat{\beta}_3 = 0, \quad SE(\hat{\beta}_3) = 2$$

Using a significance level of $\alpha = 0.05$, determine which predictors are statistically significant. State the null and alternative hypotheses for each predictor and show all steps in your calculation.

B/c this is a 2-tailed test, the significance number for z would be 1.96, and t would depend on df (n-1).

To determine if X_j is associated w/ Y, for $\alpha = .05$

$H_0: B_j = 0$ vs $H_a: B_j \neq 0$

test stat: $t = \frac{\hat{B}_j}{SE(\hat{B}_j)}$

$\hat{B}_1 = 5$, $SE(\hat{B}_1) = 1 \rightarrow t = \frac{5}{1} = 5$ H_a ✓ statistically sig $H_0: B_1 = 0$, $\boxed{H_a: B_1 \neq 0}$

$\hat{B}_2 = -3$, $SE(\hat{B}_2) = .5 \rightarrow t = \frac{-3}{.5} = -6$ H_a ✓ statistically sig $H_0: B_2 = 0$, $\boxed{H_a: B_2 \neq 0}$

$\hat{B}_3 = 0$, $SE(\hat{B}_3) = 2 \rightarrow t = \frac{0}{2} = 0$ H_0 ✓ statistically insig $\boxed{H_0: B_3 = 0}$, $H_a: B_3 \neq 0$

Problem 4: Multicollinearity in MLR (3 pts)

Briefly explain how multicollinearity can impact the interpretation of regression coefficients in multiple linear regression. Provide an example scenario where multicollinearity could occur.

Multicollinearity means one or more predictors are strongly correlated with other(s). As a result, a model can struggle to determine how significant a coefficient is to the model, and inflate coefficients.

Example: I want to predict the price of a truck based on size, engine power, towing capacity, and wheel size. Multicollinearity would happen here b/c engine power and towing capacity are highly correlated.

Problem 5: Detecting Overfitting (2 pts)

Suppose you fit two models on the same dataset:

- Model A: 3 predictors, $R^2 = 0.85$, Adjusted $R^2 = 0.82$
- Model B: 6 predictors, $R^2 = 0.88$, Adjusted $R^2 = 0.79$

Which model do you prefer and why? Support your answer using your understanding of model complexity, overfitting, and adjusted R^2.

I choose Model A. Model A has 3 predictors and will be less prone to overfitting. Although Model B does have a higher R^2, Model A's adjusted R^2 is higher, which is more important for generalization and filters out predictors that don't contribute to accuracy more.

Problem 6: Interpreting Coefficients and Making Predictions (3 pts)

You are using a model to predict house prices based on:

- Number of Bedrooms (X_1)
- Square Footage (X_2)
- Age of the House (X_3)

The fitted regression model is:

$$\text{Price} = \beta_0 + \beta_1(\text{Bedrooms}) + \beta_2(\text{Square Footage}) + \beta_3(\text{Age}) + \varepsilon$$

with estimated coefficients:

$$\hat{\beta}_0 = 50{,}000, \quad \hat{\beta}_1 = 10{,}000, \quad \hat{\beta}_2 = 150, \quad \hat{\beta}_3 = -500$$

1. Interpret each coefficient.
2. Predict the price of a house with 3 bedrooms, 2000 square feet, and 10 years old.

B_0 = Acts as a baseline (intercept). In reality, no house w/ 0 age, 0 ft², and 0 bedrooms would sell for \$50k.

B_1 = change in cost per bedroom

B_2 = change in cost per ft²

B_3 = change in cost per year aged

Price = 50000 + 10000(3) + 150(2000) − 500(10) + ε

Price = \$375000 or price = \$375000 + ε